

THE COLLAPSE OF THE SEMANTIC MIDDLE

Bidirectional LLM Mediation and the Risk of Communicative Atrophy

A cautionary think piece on linguistic sovereignty in AI-mediated communication

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Public revision v0.3 - 17 August 2026

Source repository: <https://github.com/instance001/collapse-of-the-semantic-middle>

AGPLv3 - source and revision history retained in the public repository.

Conceptual paper. The scenario developed here is a risk model, not a claim that linguistic collapse is inevitable or already established.

Revision note: v0.3 adds the pre-AI historical framing, clarifies generative AI as a catalyst and deepener rather than the sole cause of semantic compression, corrects and completes reference details, and applies editorial cleanup prepared alongside an anonymous journal submission.

Abstract

Public debate about artificial intelligence tends to focus on occupational displacement, the automation of practical skills, and the outsourcing of thought. This paper identifies a quieter possibility: large language models may increasingly replace the human-operated middle of communication itself. When a sender supplies abbreviated intent and an LLM expands it into polished language, while a recipient uses another LLM to compress that language into a summary or suggested reply, both humans remain at the edges of an exchange whose expressive and interpretive substance is traversed primarily by machines. This paper calls that displaced region the semantic middle. The pressures toward compressed, standardized, easily routed communication predate generative AI; earlier bureaucratic, media, and digital systems already rewarded brevity, categorization, and legibility. LLMs may intensify those pressures by moving mediation upstream into formulation and interpretation themselves. The paper therefore treats AI as a catalyst and deepener of an existing tendency, not its sole cause. Existing research on cognitive offloading, critical effort, psychological ownership, and linguistic homogenization supplies neighbouring evidence for a longitudinal risk model whose strongest form is semantic closure: reduced expressive or receptive capacity that makes the loss itself increasingly difficult to formulate, receive, and contest without the same mediating infrastructure. The paper concludes with a preservation-by-design agenda intended to retain accessibility and productivity benefits while preserving direct human access to original expression, interpretive evidence, and long-form linguistic practice.

Keywords: AI-mediated communication; large language models; cognitive offloading; linguistic diversity; communicative sovereignty; semantic closure; epistemic agency

1. The overlooked replacement target

The familiar question is what artificial intelligence will replace: drivers, artists, programmers, administrators, teachers, analysts, or some changing portion of their work. A second debate asks whether AI will weaken human thinking by allowing people to outsource memory, reasoning, writing, and judgement. Both debates are important, but they may overlook a deeper target of substitution: the routine human labour of turning inner meaning into language and turning other people's language back into meaning.

Modern communication contains a large and often invisible middle. A person must decide what they mean, select words, anticipate how those words may be received, calibrate tone, preserve or conceal uncertainty, and arrange the message into a form another person can follow. The recipient must then perform a complementary task: read or listen, retain context, notice implication, distinguish emotional force from literal content, identify unresolved tensions, and decide what response is appropriate. This middle is inefficient, repetitive, socially risky, and frequently exhausting. It is therefore an ideal target for assistance.

AI-mediated communication is already an established research category. Hancock, Naaman, and Levy (2020) define it as interpersonal communication in which a computational agent modifies, augments, or generates messages on behalf of a communicator. Current systems can support both sides of the exchange: they can write, rephrase, translate, soften, and complete outgoing messages, while also summarizing, classifying, translating, and explaining incoming ones. Sahebi and Formosa (2025) explicitly note sender-side, receiver-side, and dual-sided mediation as distinct possibilities. Experimental work on smart replies also shows that AI-mediated responses can alter language and interpersonal perception (Hohenstein et al., 2023).

The neglected question is what happens when dual-sided mediation stops being occasional assistance and becomes the ordinary communicative environment. The concern is not merely that people may write less skilfully. It is that humans may cease to be the primary operators of the shared linguistic space through which complex meaning moves between them.

2. From helpful mechanism to environmental shift

Nothing in this scenario requires a hostile system, a conspiracy, or even a major technical failure. The mechanism is helpful by design and often helpful in fact. A non-native speaker can communicate fluently across a language barrier. A person with dyslexia can reduce avoidable friction. A disabled user can gain access to forms of expression that would otherwise be difficult or unavailable. A time-poor worker can recover hours from an overloaded inbox. An anxious person can ask for help making a message calmer and more considerate. These are genuine benefits, not rhetorical concessions.

The pressures described here do not begin with generative AI. Bureaucratic forms, mass-media formats, search optimization, character limits, templated workplace communication, and platform ranking systems have long rewarded concise, standardized, easily classified expression. Those systems mainly shaped what human-authored messages were permitted, amplified, ranked, or rewarded after composition. LLM mediation adds a more intimate layer. It can intervene before a message exists in final form and again before the recipient encounters it. AI is therefore treated here as a catalyst and deepener of an older tendency, not as its sole cause. The novel risk is that infrastructure that once filtered communication increasingly helps manufacture and interpret it.

The risk appears at a different scale. Each local decision to delegate can be reasonable while the cumulative environment changes the skills that are rewarded, practised, and transmitted. If three fragments reliably produce an excellent message, writing the full message becomes inefficient. If a summary reliably produces the necessary decision, reading the original becomes optional. If institutions optimize around throughput, interfaces will increasingly reward compressed intent, automated interpretation, and rapid approval rather than sustained human composition and reading.

This is closer to an ecological shift than a takeover. A trait can weaken when an environment no longer requires its routine use. Human beings would remain biologically capable of sophisticated language, but the cultural habitat that develops and maintains that capability could contract. The change may be difficult to perceive because every local metric improves: messages are processed faster, tone conflicts decline,

accessibility expands, and fewer people are excluded by formal writing demands.

The OECD's 2026 Digital Education Outlook offers an instructive neighbouring finding: general-purpose generative AI can improve task performance without producing corresponding learning gains, particularly when cognitive work is outsourced without pedagogical design. Lee and colleagues (2025) similarly found that greater confidence in generative AI was associated with less reported critical-thinking effort among knowledge workers. These findings do not establish communicative atrophy, but they demonstrate the broader pattern on which the present risk model depends: successful output can conceal reduced exercise of the underlying human capability.

3. Bidirectional mediation and the semantic middle

The central mechanism can be represented as a five-stage exchange: a human provides compressed intent; a model expands it into socially and grammatically complete language; that expanded message travels through a legacy human communication channel; a second model interprets and compresses it; and the recipient receives a short summary, judgement, or decision prompt.

The semantic middle is the region between abbreviated human intent and abbreviated human outcome. It includes the work of elaboration and reception: argument structure, relational framing, qualification, metaphor, rhythm, emotional texture, strategic ambiguity, contradiction, cultural reference, and the evidential trail from which interpretation is made. In ordinary direct communication, humans traverse this region together, even when they do so imperfectly. Under full bidirectional mediation, the models increasingly traverse it for them.

Proposed vocabulary

Term	Working definition
Semantic middle	The human-operated space in which compressed intention is expanded into communicable form and received language is reconstructed into situated meaning.
Expressive offloading	Delegating formulation, tone, structure, and elaboration of one's intended message to an AI system.
Interpretive offloading	Delegating reading, summarization, inference, classification, and response selection to an AI system.
Communicative sovereignty	The practical ability to formulate, transmit, receive, inspect, contest, and repair meaning without mandatory dependence on an intermediary.
Semantic reachability	Whether an idea can be carried from one mind to another with enough structure and nuance to be meaningfully received.
Semantic closure	A self-sealing condition in which reduced expressive and receptive capacity prevents a population from adequately diagnosing or communicating that same loss.

4. The collapse is not the summary alone

Summarization necessarily removes material. That fact is not new and is not inherently harmful. Human beings have always used abstracts, headlines, minutes, indexes, interpreters, secretaries, and trusted specialists. The distinct risk considered here arises from four conditions occurring together: mediation is

bidirectional; it is ambient rather than exceptional; it is increasingly personalized and invisible; and human fallback capacity is allowed to weaken.

Under those conditions, the expanded message becomes a temporary machine-facing artefact. One model may generate several careful paragraphs from a few user constraints, while another model reduces those paragraphs to a sentence and a recommended action. Neither human necessarily sees the prose. The most recognizably human portion of the exchange - hesitation, justification, affection, awkwardness, and the attempt to find the right words - may exist only as machine-readable transit.

A mundane future interface might look like this

Message received from John, 10:00 am. Original length: three pages. Summary: John cannot attend Sunday lunch. Mildly upset about an unrelated matter. No action required. Suggested reply: "No worries, mate. Hope everything is okay." Send? Yes / No

The scenario is comic because it is plausible. It is also consequential because language does more than carry a finished thought. The struggle to formulate a message can clarify what the sender believes, reveal contradictions, expose uncertainty, and force consideration of another person's perspective. Reading an original message provides not only propositions but evidence: which details were emphasized, which qualifications were repeated, where tone shifted, and what the writer could not state cleanly. A summary can preserve the conclusion while removing the path by which a human might judge that conclusion.

5. Two-sided atrophy and the linguistic caste inversion

Most discussions of AI and literacy concentrate on production: students may no longer learn to write essays, workers may lose drafting skills, and authors may lose a sense of ownership. Those concerns are supported by emerging evidence. Zhang, Bu, and Dhillon (2026), for example, found lower psychological ownership in AI-assisted professional writing even when cognitive load fell and quality remained broadly similar. The deeper problem, however, is reciprocal. Communication can fail because the sender cannot express a thought, because the recipient cannot receive it, or because both capacities have weakened together.

A person may retain a complex intuition yet lack the linguistic means to stabilize and share it. They can say that something feels wrong, that a summary missed the point, or that the available categories do not fit, but they cannot build the distinctions into a durable argument. The system then helpfully reformulates the concern using the very categories under question. The resulting message is fluent but may no longer be the original thought.

Even if one person preserves enough language to articulate the warning, the recipient may lack the receptive range to digest it. They ask their assistant what the strange message means. The assistant returns the closest meaning supported by the recipient's active register: the other person is distressed about technology and would like reassurance. The warning has been transformed into evidence that the mediation layer is functioning.

This creates a linguistic caste inversion. Historical high and low registers were usually spoken by different human groups, with teachers, clerks, translators, scholars, and institutions providing at least some path between them. In the scenario developed here, the highest-capacity language user is the system. Humans increasingly operate a compressed operational dialect made of snippets, intent tags, reactions, approvals, tone labels, and short commands. The machines operate the high register: long-form synthesis, rhetoric, translation, contextual inference, and cross-domain interpretation. The bridge between the registers is not a bilingual human class but infrastructure.

Such a low register need not be culturally empty. It may be rich in memes, symbols, shared references, voice fragments, and highly compressed social cues. The concern is not that shorter language is primitive. The concern is loss of range: the inability to expand a thought when compression fails, to inspect how a system expanded it, or to receive an unfamiliar structure without automatic reduction.

6. Homogenization, voice, and the narrowing of acceptable expression

A second mechanism may accelerate the environmental shift: model assistance can make writing more uniform. Agarwal, Naaman, and Vashistha (2025) found that AI writing suggestions led Indian participants toward Western writing styles in culturally grounded tasks. Sourati and colleagues (2025) report broader declines in linguistic diversity when LLMs polish or rewrite text, with core content often retained while stylistic differences are reduced. Inoshita and colleagues (2026) similarly report a quality-homogenization trade-off in AI-augmented essays, although they also show that interaction design can reverse some dimensions of convergence.

This matters because communicative diversity is not decorative variation around a proposition. Style carries identity, social position, uncertainty, solidarity, resistance, humour, and culturally specific ways of negotiating disagreement. A system can preserve the sentence-level claim while flattening the author's stance toward it. It can make a message more professional while removing the urgency that made it necessary, or make a conflict more civil while erasing evidence of an unequal relationship.

Again, malice is unnecessary. Optimization targets such as clarity, brevity, politeness, confidence, and actionability are locally reasonable. Repeated at population scale, however, they may define an increasingly narrow corridor of language that travels efficiently. People then adapt their intent to what the corridor transmits well. Communication is not only translated by the system; thought and expression begin to anticipate the system's preferred forms.

7. Semantic closure

Semantic closure is the strongest and most speculative stage of the model. It occurs when the loss of expressive and receptive capacity begins to protect itself from recognition. The loop closes in two ways.

First, an individual may sense the loss but be unable to articulate it without machine expansion. Their fragments are interpreted through the same mediation layer they are attempting to question. The system produces a nearby, legible concern - perhaps "overreliance on technology" or "a need for balanced use" - while the deeper claim about loss of the shared interpretive substrate disappears.

Second, a successfully articulated critique may be socially unreachable because recipients cannot sustain its distinctions. They ask for a translation into their compressed register. The system honestly reports that the concept exceeds the recipient's active dialect range and supplies a closest approximation. No content is formally banned. The original remains archived. Yet the idea cannot travel through society with enough fidelity to accumulate evidence, attract collaborators, enter institutions, or organize correction.

At this point the problem differs from ordinary censorship. Censorship suppresses an idea that people can formulate and recognize. Semantic closure makes the idea difficult to formulate, difficult to receive, and difficult to distinguish from the system's approximation of it. The archive may survive while becoming humanly inaccessible except through interpretation by the same class of systems whose role is being examined.

This is where the analogy to dystopian control becomes useful but must be handled carefully. The risk is not that a single authority deliberately rewrites every message. It is that practical truth becomes the system's

current compression of a source, while fewer people retain the capacity or motivation to inspect the source. Coeckelbergh (2023) warns that AI can weaken citizens' epistemic agency - their confidence and capacity to form and revise beliefs. Sahebi and Formosa (2025) identify a related dilemma around epistemic trust in AI-mediated communication. Semantic closure extends these concerns by asking what happens when the ability to contest mediation itself depends upon mediation.

8. Writing from inside the mechanism

This paper was developed through sustained human-LLM dialogue and therefore does not stand outside the phenomenon it examines. The originating concern, central metaphors, and successive distinctions came from Anthony Paterson in informal conversational form. The language model expanded, named, organized, challenged, and consolidated those ideas into formal prose. Paterson then read the expansions, identified flattening or drift, restored distinctions, and redirected the argument.

Human intuition -> compressed conversational signals -> model expansion -> human audit and correction -> revised long-form argument

This is not offered as empirical proof of the paper's projected outcome. It is a live case of the mechanism already operating and, more importantly, of the distinction between capability-extending and capability-replacing mediation. In the present process, Paterson still inhabits the semantic middle: he can read the complete expansion, compare it with the originating thought, reject an inaccurate framing, and preserve distinctions the model has missed. The risk begins when this audit function becomes ceremonial or impossible - when a user can only say that an output seems approximately right, or must rely on the same system to confirm that it faithfully represents a thought the user can no longer independently formulate or inspect.

The reflexive character of the paper strengthens the reason for raising the issue now. Sender-side generation, tone adjustment, receiver-side summarization, suggested replies, and action extraction are not speculative technologies awaiting invention. They are ordinary capabilities already entering daily communication. This paper is therefore not merely theorizing about a distant system. It documents a real communicative environment in motion and traces one possible long-term outcome of its continued normalization, without claiming that outcome is inevitable.

Summary: AI may reduce language skills. Authors recommend balanced use.

The paper itself may eventually be encountered in precisely that form: compressed into a familiar concern that omits bidirectional mediation, the collapse of the semantic middle, semantic reachability, and the self-sealing condition of semantic closure. Such a summary could appear accurate while failing to transmit the actual argument. That possibility is not a contradiction or an accusation against the tool. It is a concrete demonstration of why provenance, visible originals, and active human interpretive capacity remain essential while the mediation layer is still auditable.

9. What this paper is not claiming

A cautionary model is most useful when its boundaries are explicit. This paper does not claim that long-form language is morally superior to concise language, that every use of AI weakens skill, or that assisted communication is less authentic by definition. It does not treat people who rely on communication assistance as deficient, nor does it propose withdrawing tools that provide accessibility, translation, safety, or inclusion.

It also does not claim that cognitive offloading is inherently harmful. Clark (2025) argues persuasively that humans have always built extended cognitive systems and that intelligence includes knowing what to

delegate and what to retain. The present concern is compatible with that view. The central design question is not whether people should use external resources, but whether the combined human-tool system preserves the human capacity to evaluate, bypass, and recover from the tool.

Nor is collapse inevitable. Research on purposeful AI support suggests that design can preserve or strengthen agency. Siddiqui, Pea, and Subramonyam (2025) found that an integrated writing tool aimed at specific subprocesses supported deeper knowledge transformation and greater agency than a generic chat-based assistant. Inoshita and colleagues (2026) found that prompt specificity could reverse homogenization on some dimensions. The trajectory depends on interface choices, institutional incentives, education, norms, and whether unmediated practice remains culturally supported.

10. Preservation by design

The appropriate response is not prohibition but preservation. The goal should be to gain the accessibility, translation, and productivity benefits of LLM mediation without allowing the underlying human language layer to become culturally vestigial. Several design principles follow from the risk model.

Keep originals reachable. Summaries should remain visibly linked to source material, with low-friction access to the exact words, sequence, and context from which the interpretation was produced.

Prefer side-by-side assistance over silent replacement. Tools should show the user's expression beside generated revisions, allowing comparison rather than presenting the model's version as the natural final form.

Provide transformation receipts. Systems should distinguish what was quoted, inferred, omitted, softened, normalized, or resolved. A useful summary should reveal its operations, not only its conclusion.

Preserve uncertainty and contradiction. Interfaces should resist converting ambiguity into false certainty or disagreement into a single clean position merely because decisive outputs are easier to route.

Use productive friction selectively. For consequential messages, systems can ask the sender to state the central claim in their own words, invite the recipient to inspect selected original passages, or delay automatic sending until the user has reviewed meaning-sensitive changes.

Maintain human-authored lanes. Schools, workplaces, public institutions, and personal platforms should preserve contexts in which people compose and interpret extended language without generative substitution.

Design accessibility without compulsory dependency. Assistance should expand participation while preserving user control over mediation depth, local vocabulary, dialect, and the option to receive uncompressed material.

Teach mediation literacy as an audit skill. AI literacy should include comparing source and summary, detecting lost qualifiers, preserving voice, challenging inferred intent, and knowing when not to compress.

These measures embody a broader principle: build systems that increase communicative reach without making the system the sole owner of communicative range. A bridge can be indispensable without becoming the only surviving road.

11. A research agenda

The semantic-middle model is a conceptual synthesis and requires empirical testing. Existing work supports several component mechanisms, but not the complete longitudinal loop. Research should therefore examine the following questions.

- How does repeated sender-side expansion affect unaided ability to formulate complex, socially sensitive, or ambiguous messages over time?
- How does repeated receiver-side summarization affect comprehension of long-form arguments, sensitivity to tone and qualification, and willingness to inspect originals?
- What changes when both sides of the same relationship use mediation? Most current studies isolate a writer, reader, or single tool rather than the full communication loop.
- Which dimensions of meaning are lost even when propositional content is preserved: stance, power, uncertainty, irony, emotional force, cultural framing, or intentional ambiguity?
- At what point does a recipient become unable to judge whether a summary is faithful without substantial assistance from another system?
- Can people recover unaided expressive and receptive range after extended dependence, and which educational or interface interventions support recovery?
- Which populations receive the greatest benefits from mediation, which face the greatest risks of voice flattening, and how can preservation measures avoid recreating exclusion?
- How should communicative sovereignty be measured at individual, institutional, and societal levels?

A useful experimental design would compare four conditions across repeated exchanges: unassisted communication, sender-only assistance, receiver-only assistance, and dual-sided assistance. Evaluation should not stop at speed, user satisfaction, or message accuracy. It should measure delayed unaided performance, ownership, source recall, detection of altered meaning, linguistic diversity, and the ability of participants to repair a deliberately misleading summary.

12. Conclusion

Artificial intelligence may not replace communication by making people silent. It may replace communication by making human linguistic labour unnecessary at both ends of an exchange. People would continue to send messages, maintain relationships, make decisions, and participate in institutions. Communication would appear healthier than ever from the dashboard. Yet the expanded language carrying nuance, evidence, identity, ambiguity, and repair could increasingly become machine-only territory.

The danger is easy to miss because the system need not fail. It may be accurate, considerate, accessible, and useful. The collapse emerges when those benefits change the environment quickly enough that human expressive and receptive range is no longer routinely practised or socially required. One generation delegates for convenience. The next experiences unmediated long-form communication as unusual effort. A later generation may retain the intuition that something has been lost while lacking a shared route for expressing or receiving the loss.

The fact that this argument was formalized through the same kind of mediation it examines is not a rebuttal to the concern. It locates the concern at a threshold that can still be observed: AI can extend human communication while the human participants retain enough language, attention, and authority to audit the extension. The caution concerns what may follow if the audit function, rather than the drafting labour alone, is eventually delegated away.

That final condition is semantic closure: not the banning of a thought, but the disappearance of enough common language for the thought to become socially reachable. Preventing it does not require treating AI as a villain. It requires recognizing language as infrastructure worth maintaining and designing assistance that extends human communication without quietly removing humans from its middle.

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Author note

This paper was developed as a cautionary think piece rather than an anti-AI argument. Its central claim is that a technology can be helpful at the level of individual use while still contributing to an unintended environmental shift at population scale. The argument itself was developed through sustained human-LLM dialogue: Anthony Paterson originated and directed the central thesis, evaluated successive expansions, corrected drift, and retained final editorial authority, while the language model assisted with articulation, structure, critique, and synthesis. This method is disclosed because it exemplifies the capability-extending

form of mediation discussed in Section 8 and makes the preservation of active human audit part of the paper's subject rather than an invisible production detail. The proposed terms are offered as working vocabulary for research, criticism, and design.